

DETERMINANTS OF STOCK MARKET DEVELOPMENT

Team Members-

- 1) Aneesh Nitin Bhate (b.aneesh@iitg.ac.in)
- 2) Aashish Pendharkar (p.aashish@iitg.ac.in)
- 3) Arnav Jain (j.arnav@iitg.ac.in)
- 4) Aryan Tripathi (aryan.tripathi@iitg.ac.in)
- 5) Arnav Tripathi (arnavt@iitg.ac.in)
- 6) Atharva Agrawal (atharva.agrawal@iitg.ac.in)
- 7) Sanjay Saini (sanjay.saini@iitg.ac.in)

Academic Year 2025-2026
HS163: Basic Econometrics

Introduction-

Investing money is a common way for people to increase their wealth over time. Many financial options are available to investors, including bonds, stocks, and options. Each of these carries different levels of risk and return. Additionally, there are more passive choices like fixed deposits, which usually offer lower returns but come with lower risk.

Participation in financial markets often reflects how willing a population is to take risks. It also shows their overall financial stability and flexibility. Higher participation may suggest greater confidence in the economy and easier access to financial resources.

Through this study, we will look at the relationship between various economic and financial factors and their impact on the total value of stocks traded in a country. By understanding these connections, we hope to gain insights into how different conditions affect market activity and investor behavior.

The study's results can help create practical policy suggestions that encourage more people to participate in financial markets. By pinpointing the main factors that affect how investors behave, policymakers can develop strategies that make investing easier, clearer, and more appealing to a broader audience. More people participating in markets can lead to better resource distribution, increased liquidity, and stronger capital development in the economy. In the long run, this can support business growth, innovation, and job creation. As more individuals get involved with financial markets, it can also boost financial understanding and help build long-term wealth, ultimately aiding the overall economic growth and prosperity of a nation.

Data Set Utilised-

The dataset used in this study comes from the World Bank DataBank, one of the largest publicly available sources of global data. It gathers information from many surveys and research projects that cover key socio-economic indicators across countries.

For this study, we focused on data from the World Development Indicators. This source offers reliable and consistent statistics on economic performance, financial activity, and development metrics. It is known for its dependability and ability to compare data, which makes it useful for examining trends and relationships between different nations.

There were 24 countries under consideration for this study - India, Argentina, Austria, Australia, United Kingdom, United Arab Emirates, United States, Mexico, Germany, France, Egypt, China, Japan, Italy, Malaysia, New Zealand, Norway, Oman, Russia, Singapore, Switzerland, South Africa, Saudi Arabia, Canada.

stock_value: This is the main metric used in the study. It refers to the total value of stocks traded, measured in current US dollars. It represents the overall value of all shares traded on domestic exchanges within a country over a specific period. By focusing on value rather than volume, this metric captures the overall scale of market activity and liquidity. Standardizing the data in US dollars allows for consistent comparisons across countries with different currencies and economic sizes.

adj_income: Adjusted net national income per capita reflects the average income of a country's citizens after accounting for factors like the depreciation of physical capital and the depletion of natural resources. Unlike gross national income per capita, this measure gives a clearer picture of sustainable income and long-term economic well-being, as it considers the loss of national wealth due to resource use and asset wear.

taxes_income: This metric shows the level of taxes imposed by the government on income, profits, and capital gains, expressed as a percentage. It indicates the financial burden on individuals and businesses and can affect investment decisions. Higher taxes may discourage participation in financial markets, while moderate and well-structured tax systems can encourage compliance and stable economic activity.

listed_companies: This refers to the total number of domestically listed companies on a country's stock exchanges. It serves as an indicator of market depth and development. A higher number of listed firms usually reflects a more developed financial market with more opportunities for investment and diversification.

mobile_subs: The number of cellular subscriptions per country includes all active mobile connections across service providers. This metric is often used as a proxy for technological access and connectivity. More access to mobile networks can support financial inclusion, improve access to market information, and enable participation in digital trading platforms.

employment_ratio: This represents the proportion of the working-age population (aged 15 and above) that is currently employed. It is a key indicator of labor market health and economic productivity. A higher employment ratio suggests greater income stability and purchasing power among the population, which can influence the level of participation in financial markets.

Methodology-

1. Data Cleaning

The dataset initially contained several issues that needed to be resolved before it could be used for analysis. Some columns had missing values, while others contained extremely large values that required transformation for easier handling and interpretation. In addition, certain variables were stored as strings instead of numeric types, so they were converted into appropriate numerical formats. Column names were also standardized and simplified for consistency and ease of use.

Dropping Unnecessary Columns

The column TimeCode was removed as it was redundant, since the dataset already included a year column conveying the same information. The data was then sorted by country_code and year to ensure a clear and structured arrangement, making it easier to analyze trends over time.

Imputing Missing Data Points

To manage missing values effectively, we applied different imputation techniques based on the type and behavior of each variable. The goal was to keep the integrity of the data while making reasonable and informed estimates.

- Manual imputation -In cases where missing values represented real-world conditions instead of data gaps, we filled values directly based on known facts. For example, countries like Oman and the UAE do not impose capital gains tax, so missing entries in taxes_income were set to 0. This ensures the data reflects policy rather than treating it as missing information.
- Interpolation -For continuous variables such as mobile_subs and employment_ratio, we used interpolation. This method estimates missing values by finding trends between known data points over time and filling in gaps accordingly. It is particularly useful for time series data because it maintains the natural progression and avoids sudden changes. For missing values at the beginning or end of a series, we used the nearest available values to maintain continuity.

- Regression imputation -For variables like taxes_income, we applied regression imputation. This involves creating a statistical model using related variables such as adj_income and employment_ratio to predict missing values. By using relationships between variables, this method provides more informed estimates compared to simpler techniques like mean substitution.
- Median imputation -For listed_companies, we used median imputation. This technique replaces missing values with the median of the available data. It is especially helpful when the data may contain outliers since the median is less affected by extreme values and offers a more reliable central estimate.

2. Regression Model-

$$\log(stock_value) = \beta_0 + \beta_1 \log(adj_income) + \beta_2 employment_ratio + \beta_3 mobile_subs + \beta_4 taxes_income + \beta_5 listed_companies + \varepsilon$$

Ordinary Least Squares (OLS) regression is a statistical method used to estimate the relationship between a dependent variable and one or more independent variables. It fits a linear equation to observed data by minimizing the sum of squared differences between actual and predicted values, known as residuals.

OLS assumes a linear relationship between variables and, under the Gauss Markov assumptions such as absence of perfect multicollinearity, constant variance of errors, and no autocorrelation, it produces the Best Linear Unbiased Estimates. Due to its simplicity and interpretability, OLS is widely applied in economics, finance, and social sciences to analyze relationships, make predictions, and evaluate the impact of various factors. Common applications include economic growth analysis, financial market studies, pricing models, demand estimation, and policy evaluation.

Reasons for choosing OLS regression

Linearity assumption

OLS is based on the assumption of a linear relationship between the dependent variable and independent variables. In this study, the relationship between market activity and key economic indicators is expected to be approximately linear, making OLS an appropriate initial approach.

Interpretability of results

OLS provides coefficient estimates that clearly indicate the direction and magnitude of the relationship between variables. This allows for straightforward interpretation of how each independent variable influences the dependent variable.

Multiple predictors

OLS supports the inclusion of multiple independent variables in a single model. Since market activity is influenced by several factors such as income levels, taxation, employment, and technological access, a multivariate framework is necessary.

Statistical significance testing

OLS enables hypothesis testing through methods such as t tests and F tests. These tests help determine whether the relationships observed are statistically significant and identify the most relevant predictors.

Availability of large datasets

The dataset used, sourced from the World Bank, specifically the World Development Indicators, provides a large number of observations across countries and years. This improves the reliability and robustness of the estimated results.

Procedure-

After loading the raw data set into Stata, we first clean the data set using the Cleaning command set as specified in the code snippet below, this goes through the above mentioned procedures and prepares the data for applying the model to it.

```
* 1. Convert "." to missing
foreach var of varlist _all {
    capture confirm string variable `var'
    if !_rc {
        replace `var' = "" if `var' == "."
    }
}

* 2. Destring numeric variables
destring Stockstradedtotalvaluecurr, replace ignore("")
destring Adjustednetnationalincomeper, replace ignore("")
destring Taxesonincomeprofitsandcap, replace ignore("")
destring Listeddomesticcompaniestotal, replace ignore("")
destring MobilecellularsubscriptionsI, replace ignore("")
destring Employmenttopopulationratio, replace ignore("")

* 3. Rename variables (make them readable)
rename CountryName country
rename CountryCode country_code
rename Time year

rename Stockstradedtotalvaluecurr stock_value
rename Adjustednetnationalincomeper adj_income
rename Taxesonincomeprofitsandcap taxes_income
rename Listeddomesticcompaniestotal listed_companies
rename MobilecellularsubscriptionsI mobile_subs
rename Employmenttopopulationratio employment_ratio

* 4. Drop unnecessary variable
drop TimeCode

* 5. Order variables neatly
order country country_code year

* 6. Sort data
sort country year

* 7. Optional: Handle duplicates
duplicates drop country year, force
```



```

* 8. Manually setting values
replace taxes_income = 0 if country == "Oman"
replace taxes_income = 0 if country == "United Arab Emirates"

* 9. Interpolation for smooth variables
foreach var in employment_ratio mobile_subs adj_income {
    by country: ipolate `var' year, gen(temp)
    replace `var' = temp if missing(`var')
    drop temp
}

* 10. Filling edge values
foreach var in employment_ratio mobile_subs adj_income {
    by country: replace `var' = `var'[_n-1] if missing(`var')
    by country: replace `var' = `var'[_n+1] if missing(`var')
}

* 11. Regression imputation
reg stock_value adj_income employment_ratio mobile_subs
predict stock_hat
replace stock_value = stock_hat if missing(stock_value)
drop stock_hat

reg taxes_income adj_income employment_ratio
predict tax_hat
replace taxes_income = tax_hat if missing(taxes_income)
drop tax_hat

* 12. Median Imputation
bysort country: egen med_listed = median(listed_companies)
replace listed_companies = med_listed if missing(listed_companies)
drop med_listed

```

We can then save the cleaned data set and then apply the Processing file to perform our statistical analyses to extract information regarding the variables and the relation between them.

*Calculating Log Values for stock_value and adjusted income

```

replace stock_value = abs(stock_value)
gen log_stock= log(stock_value)
gen log_income = log(adj_income)

```

A logarithmic transformation was applied to the stock_value and adj_income variables because they contain very large and highly skewed values. In their original form, a few extreme observations could dominate the analysis and distort results.

The log transformation reduces skewness and brings the data closer to a normal distribution, improving the stability of models such as OLS. It also limits the impact of outliers and allows coefficients to be interpreted in relative or percentage terms, making comparisons across countries more meaningful.

We then initially regress on the variables assuming homoscedasticity

**Regressing assuming homoscedasticity*
`reg log_stock log_income employment_ratio mobile_subs taxes_income listed_companies`

We begin by running an initial regression under the assumption of homoscedasticity, where the variance of the error terms is constant across observations. This provides baseline estimates of the relationships between variables and serves as a starting point for further diagnostic testing and model refinement.

. reg log_stock log_income employment_ratio mobile_subs taxes_income listed_companies						
Source	SS	df	MS	Number of obs	=	600
Model	1549.83559	5	309.967117	F(5, 594)	=	97.58
Residual	1886.81135	594	3.17645009	Prob > F	=	0.0000
				R-squared	=	0.4510
				Adj R-squared	=	0.4464
Total	3436.64694	599	5.73730708	Root MSE	=	1.7823

log_stock	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
log_income	.9320131	.0752121	12.39	0.000	.7842992	1.079727
employment_ratio	-.0630166	.0100502	-6.27	0.000	-.0827549	-.0432784
mobile_subs	2.41e-09	3.77e-10	6.38	0.000	1.67e-09	3.15e-09
taxes_income	.0094582	.0048454	1.95	0.051	-.000058	.0189743
listed_companies	.0004921	.0000614	8.01	0.000	.0003714	.0006127
_cons	19.94714	.729547	27.34	0.000	18.51434	21.37995

After estimating the initial regression, we test for heteroskedasticity to assess whether the assumption of constant error variance holds. This step helps verify the reliability of the model results, as the presence of heteroskedasticity can affect the accuracy of standard errors and statistical inference.

```
.
. *Checking for heteroscedascity
. estat hettest
```

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity

Assumption: Normal error terms

Variable: Fitted values of log_stock

H0: Constant variance

```
      chi2(1) =    4.38
Prob > chi2 = 0.0364
```

With a chi square value of 4.38 and a p value of 0.0364, the null hypothesis of homoscedasticity is rejected at conventional significance levels. This indicates the presence of heteroskedasticity in the model, meaning the variance of the error terms is not constant. As a result, the regression must be adjusted to account for heteroskedasticity in order to obtain more reliable standard errors and accurate inference.

```
*Regressing with heteroscedascity
reg log_stock log_income employment_ratio mobile_subs taxes_income listed_companies,robust
```

Regressing again with heteroskedascity.

```
Linear regression              Number of obs   =          600
                              F(4, 594)         =          .
                              Prob > F           =          .
                              R-squared           =         0.4510
                              Root MSE        =         1.7823
```

log_stock	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
log_income	.9320131	.0799689	11.65	0.000	.7749568	1.089069
employment_ratio	-.0630166	.010587	-5.95	0.000	-.0838092	-.0422241
mobile_subs	2.41e-09	3.68e-10	6.55	0.000	1.69e-09	3.13e-09
taxes_income	.0094582	.0058214	1.62	0.105	-.0019749	.0208912
listed_companies	.0004921	.000068	7.23	0.000	.0003585	.0006257
_cons	19.94714	.8312428	24.00	0.000	18.31461	21.57967

We can now use these for a global understanding of the variation of stock value traded against the factors we have considered. We also calculate the VIF to check for multicollinearity in our model.

Variance Inflation Factor (VIF) is a measure used to detect multicollinearity in a regression model. Multicollinearity occurs when independent variables are highly correlated with each other, which can make coefficient estimates unstable and difficult to interpret.

```
. *Calculating Variance Inflation Factor  
. estat vif
```

Variable	VIF	1/VIF
listed_com~s	2.43	0.411669
mobile_subs	2.33	0.428463
log_income	1.43	0.699176
taxes_income	1.35	0.740806
employment~o	1.33	0.751460
Mean VIF	1.77	

From the results, all variables have VIF values below 5, indicating that multicollinearity is not a concern in the model. This suggests that the independent variables are not highly correlated with one another, allowing the coefficient estimates to remain stable and reliable for interpretation.

In general, a VIF value between 1 and 5 is considered acceptable and indicates low to moderate correlation. Values between 5 and 10 suggest increasing levels of correlation that may require attention, while values above 10 indicate a high degree of multicollinearity. In such cases, it is often advisable to remove or modify the affected variables to improve the robustness of the model.

We also calculate the residuals and plot them.

```

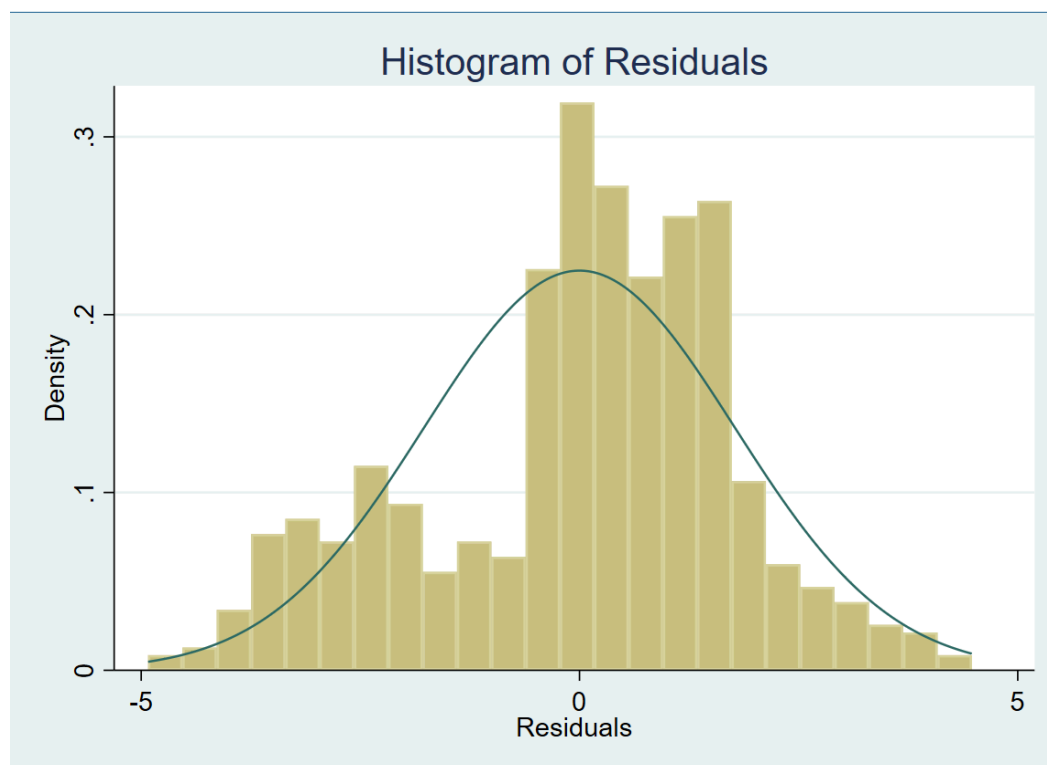
. *Calculating residuals
. predict res,residuals

.
. *Plotting Residuals
. histogram res, normal title("Histogram of Residuals")
(bin=24, start=-4.9208503, width=.39153161)

```

Residuals were calculated as the difference between the observed and predicted values obtained from the regression model. Analyzing these residuals is important for assessing the overall fit and validity of the model.

A histogram of the residuals was plotted to examine their distribution. One of the key assumptions of OLS regression is that the residuals are approximately normally distributed. If the distribution follows a bell shaped pattern, it indicates that the model is reasonably well specified and that statistical inferences drawn from it are reliable.



The histogram of residuals shows that the error terms are approximately centered around zero, which is consistent with expectations from a well specified regression model. The overall shape resembles a bell shaped distribution, suggesting that the normality assumption of OLS is reasonably satisfied.

However, there are slight deviations from perfect symmetry, with some spread on both tails, indicating the presence of mild skewness or a few extreme values. While these deviations are not severe, they suggest that the model may not capture all underlying variation in the data.

Overall, the residual distribution supports the validity of the model assumptions to a reasonable extent, and the regression results can be considered reliable, although minor improvements or robustness checks may further strengthen the analysis.

Results and Conclusions-

Linear regression

Number of obs = 600
F(4, 594) = .
Prob > F = .
R-squared = 0.4510
Root MSE = 1.7823

log_stock	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
log_income	.9320131	.0799689	11.65	0.000	.7749568	1.089069
employment_ratio	-.0630166	.010587	-5.95	0.000	-.0838092	-.0422241
mobile_subs	2.41e-09	3.68e-10	6.55	0.000	1.69e-09	3.13e-09
taxes_income	.0094582	.0058214	1.62	0.105	-.0019749	.0208912
listed_companies	.0004921	.000068	7.23	0.000	.0003585	.0006257
_cons	19.94714	.8312428	24.00	0.000	18.31461	21.57967

We can interpret the following information from the results of the Regression regarding the parameters and the effect they possess on the stock_value traded.

- **Model fit (R squared = 0.451)**

The model explains approximately 45.1 percent of the variation in stock market activity across countries and time. This indicates a moderate level of explanatory power, suggesting that while the included variables capture important determinants of stock value traded, there are still other factors not included in the model that may also influence market activity. The use of robust standard errors ensures that the estimates remain reliable even in the presence of heteroskedasticity.

- **log_income**

The coefficient for log_income is positive and highly statistically significant. Since both the dependent and this independent variable are in logarithmic form, the coefficient can be interpreted as an elasticity. Specifically, a 1 percent increase in adjusted income per capita is associated with approximately a 0.93 percent increase in stock value traded, holding other factors constant. This highlights income as a key driver of financial market participation, as higher income levels generally increase the capacity and willingness of individuals to invest.

- **employment_ratio**

The coefficient for employment_ratio is negative and statistically significant. This suggests that higher employment levels are associated with lower stock

market activity, holding other variables constant. While this may seem counterintuitive, it could reflect structural characteristics of certain economies, such as a higher proportion of employment in sectors with limited financial market engagement, or a preference for stable income over investment in financial assets.

- **mobile_sub**

The coefficient for mobile_sub is positive and statistically significant, indicating that an increase in mobile subscriptions is associated with higher stock market activity. Although the coefficient is numerically small due to the scale of the variable, the statistical significance suggests a meaningful relationship. This supports the idea that greater technological penetration and connectivity improve access to financial information and trading platforms, thereby facilitating market participation.

- **taxes_income**

The coefficient for taxes_income is positive but not statistically significant at conventional levels. This indicates that, within the scope of this model, taxation on income, profits, or capital gains does not have a strong or consistent impact on stock market activity. The lack of significance suggests that the effect of taxation may be more complex, potentially influenced by other institutional or economic factors not captured here.

- **listed_companies**

The coefficient for listed_companies is positive and highly statistically significant. This implies that an increase in the number of domestically listed companies is associated with higher stock trading value. It reflects the importance of market depth and development, as a larger number of listed firms typically provides more investment opportunities and enhances overall market liquidity.

Policy Changes and Implementations

Based on the regression results, policy recommendations can be designed around the key drivers of stock market activity identified in the study, namely income levels, technological access, and market depth. A structured approach to implementation is outlined below.

Improving income levels and financial capacity

Since income shows a strong positive relationship with market participation, policies should focus on increasing disposable income and encouraging savings that can be channeled into financial markets. Governments can promote this through tax incentives on long term investments, such as reduced capital gains tax for equities held beyond a certain period. Additionally, introducing subsidized or tax advantaged investment accounts for middle income households can encourage first time participation.

In terms of implementation, this would require coordination between fiscal authorities and financial regulators to design incentive schemes, followed by public awareness campaigns to ensure adoption. Over time, this can be complemented by policies that promote stable income growth, such as skill development programs and formal sector job creation.

Enhancing financial literacy and investor awareness

Even with sufficient income, participation may remain low if individuals lack understanding of financial markets. National level financial literacy programs should be introduced, focusing on basic investment concepts, risk management, and long term wealth creation. These can be integrated into school curricula as well as delivered through digital platforms and public campaigns.

A phased implementation plan could begin with pilot programs in urban areas, followed by expansion to semi urban and rural regions. Partnerships with financial institutions and educational bodies can improve reach and effectiveness.

Expanding digital and technological access

The positive impact of mobile penetration highlights the importance of technology in enabling market access. Policies should aim to expand affordable internet access and

smartphone usage, particularly in underserved regions. In parallel, governments can support the development of secure and user friendly digital trading platforms with simplified onboarding processes.

Implementation would involve investment in digital infrastructure, incentives for telecom providers to expand coverage, and regulatory support for fintech innovations. Ensuring strong cybersecurity frameworks is also essential to build trust among new investors.

Strengthening market depth and participation

The number of listed companies significantly influences market activity, suggesting the need to deepen capital markets. Governments and regulators can encourage more firms, especially small and medium enterprises, to list on stock exchanges by simplifying listing requirements, reducing compliance costs, and providing advisory support.

A practical approach would involve creating dedicated SME exchanges or segments with relaxed norms, along with capacity building programs to prepare firms for listing. Incentives such as tax benefits for newly listed companies can further encourage participation.

Revisiting taxation policies

Although taxes_income was not statistically significant in the model, taxation still plays an important role in shaping investor sentiment. Policymakers should aim for a balanced tax regime that does not discourage participation. This could include simplifying tax structures, ensuring clarity in capital gains taxation, and avoiding frequent policy changes that create uncertainty.

Implementation would require periodic review of tax policies in consultation with market participants, along with transparent communication to maintain investor confidence.

Improving labor market integration with financial systems

The negative relationship with employment suggests a need to better connect working populations with financial markets. Policies can promote automatic enrollment in

retirement or investment schemes linked to employment, encouraging individuals to allocate a portion of their income to market instruments.

This can be implemented through employer based programs, where contributions are either voluntary or matched by employers or the government. Over time, such systems can significantly increase participation rates.

Overall plan of action

In the short term, focus should be placed on financial literacy, digital access, and simplifying market entry barriers. In the medium term, policies should aim to expand market depth by encouraging more firms to list and improving institutional frameworks. In the long term, sustained income growth and integration of financial systems with everyday economic activity will be key to building a robust and inclusive investment ecosystem.

Together, these measures can increase participation in financial markets, improve capital allocation, and contribute to broader economic growth and stability.

Possible Improvements in Study-

Inclusion of additional variables

A broader set of variables such as interest rates, inflation, financial literacy, and institutional quality could be included to better capture the factors influencing stock market activity. This would help reduce omitted variable bias and improve the explanatory power of the model.

Extending the time horizon

Using a longer time period would allow for a more comprehensive analysis of trends and capture different economic cycles. This would improve the robustness and reliability of the results.

Use of panel data techniques

Applying panel data models such as fixed effects or random effects would help control for unobserved country specific characteristics. This would lead to more accurate estimates by accounting for inherent differences across countries.

Addressing endogeneity

Some variables may be jointly determined with stock market activity, which can bias results. Using methods like instrumental variable regression would help obtain more consistent and reliable estimates.

Incorporating dynamic effects

Including lagged variables or using dynamic models would allow the analysis to capture delayed effects of economic factors on stock market activity, providing a more realistic representation of relationships over time.

References

1. <https://databank.worldbank.org/source/world-development-indicators> (World Bank World Development Indicators- Primary Data Source)